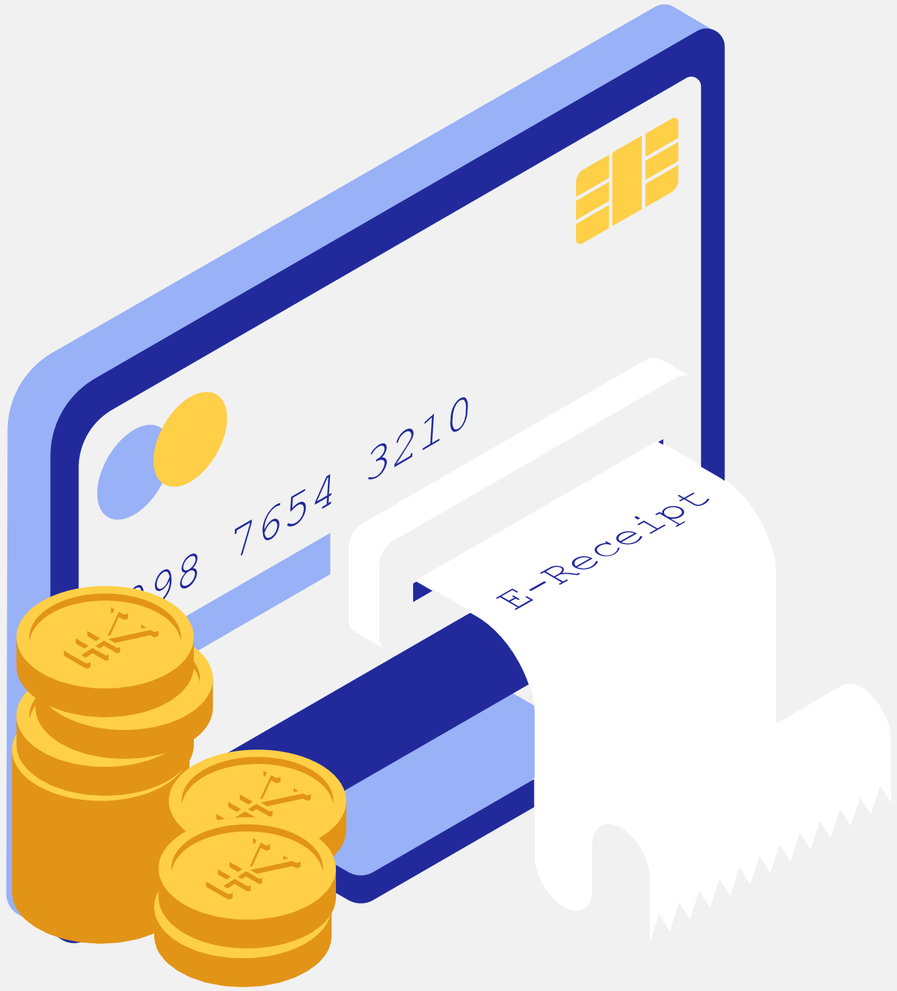

Predicting Online Purchase Intention Using Machine Learning

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Business Problem

- In e-commerce, the majority of website visitors ***do not*** complete a purchase.
- So, companies need a way to identify which user sessions are likely to result in revenue.

Solution

- We build a machine learning model that predicts the **probability of purchase for each active session**.
- These predictions help businesses identify users with **higher or lower purchase likelihood**, enabling more targeted marketing and better allocation of promotional resources.

Understanding our Data

- The dataset contains 12,330 **unique shopping sessions**, resulting in **conversion / no conversion**
- This is our target - **Revenue**

Website Interaction Features

These features describe **how users interact with the website** during a session.

- **Administrative** pages and duration
- **Informational** pages and duration
- **Product-Related** pages and duration
- **Bounce Rate** (single page sessions)
- **Exit Rate**
- **Page Value** (in $)

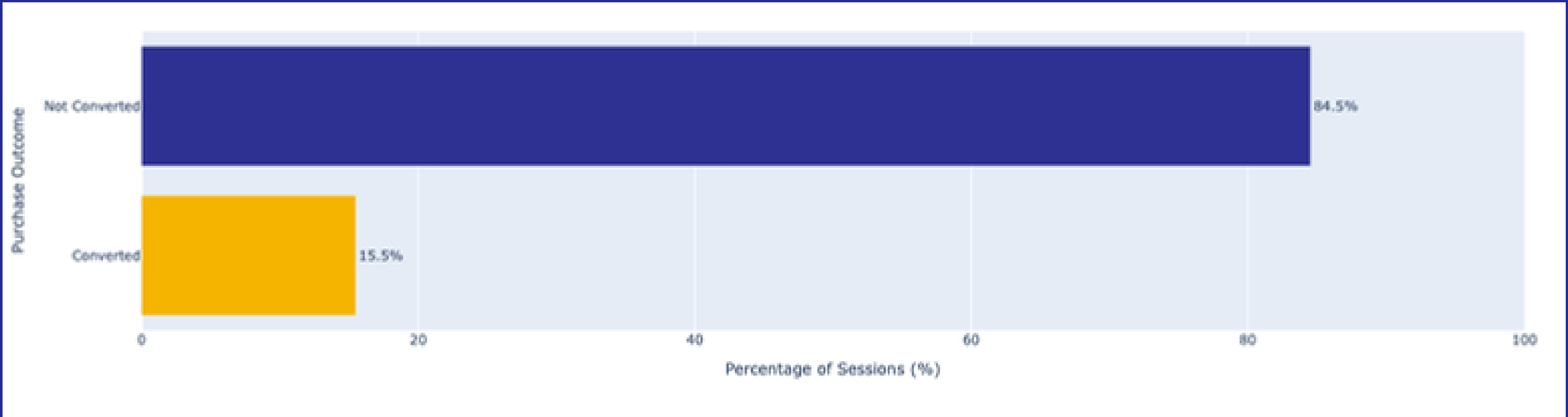
Visitor & Session Information

These features provide **contextual and demographic information** about the session.

- **Visitor Type** (New / Returning)
- **Special Day** (special occasions / holidays)
- **Month** of visit
- **Weekend** (Yes/No)
- **Operating System***
- **Browser***
- **Region** (location)*
- **Traffic Type***

**Anonymized / encoded data*

Key Challenge



High Class Imbalance
Revenue = False → 84.53%
Revenue = True → 15.47%

- **Because purchases are rare, a model predicting "No Purchase" for every session could still show high accuracy, making accuracy alone misleading.**
- **Therefore, the model should focus on correctly identifying purchase sessions rather than maximizing overall accuracy**

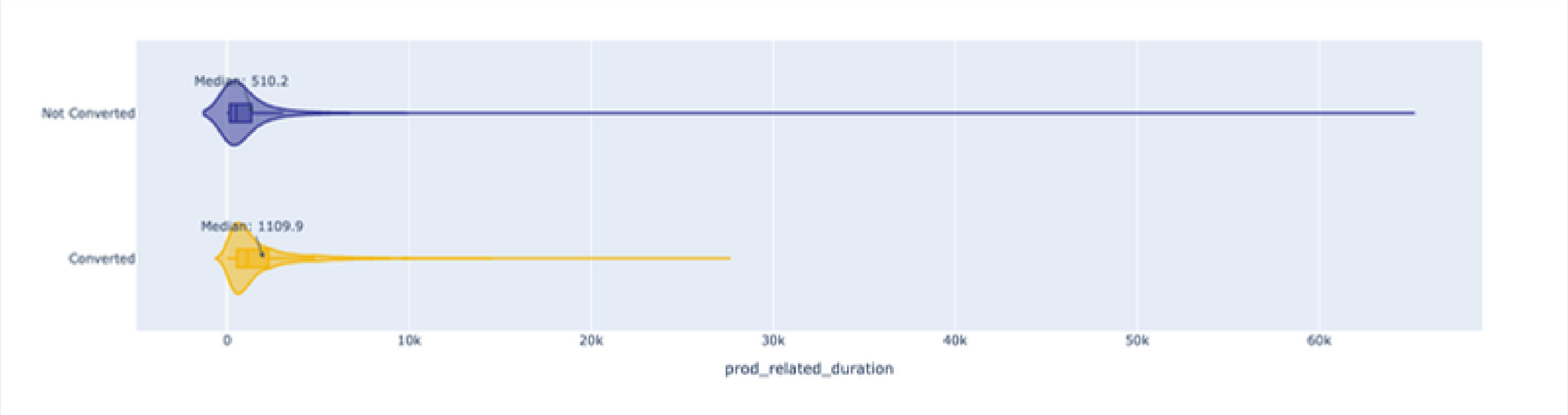
Page Value vs Revenue

- Not Converted → Median ≈ 0
- Converted → Median ≈ 16.76

- The median values show that the most **non-buyers sessions have almost zero page value** while buyers have significantly higher page values (validated using a t-test).

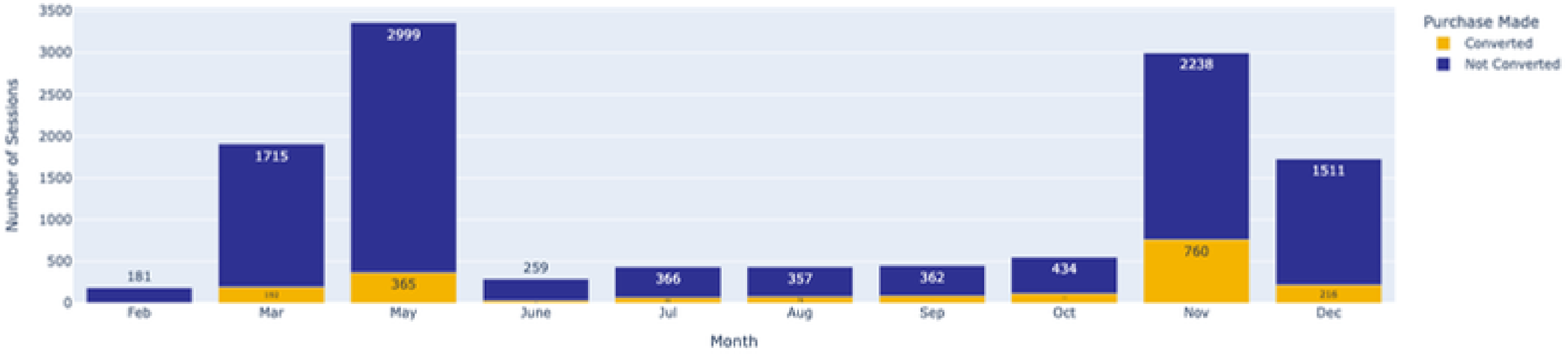
- Page Value is critical for predicting purchases, but meaningful predictions are made only when combined with other user behavior signals

Product page Duration vs Revenue



- Median values indicate that the **buyers clearly spend more time on product pages** than the non-buyers.
- The extreme values in duration, as high as 67k, as mentioned in the previous findings, could indicate casual browsing (more casual browsing = longer the duration)

Month vs Revenue



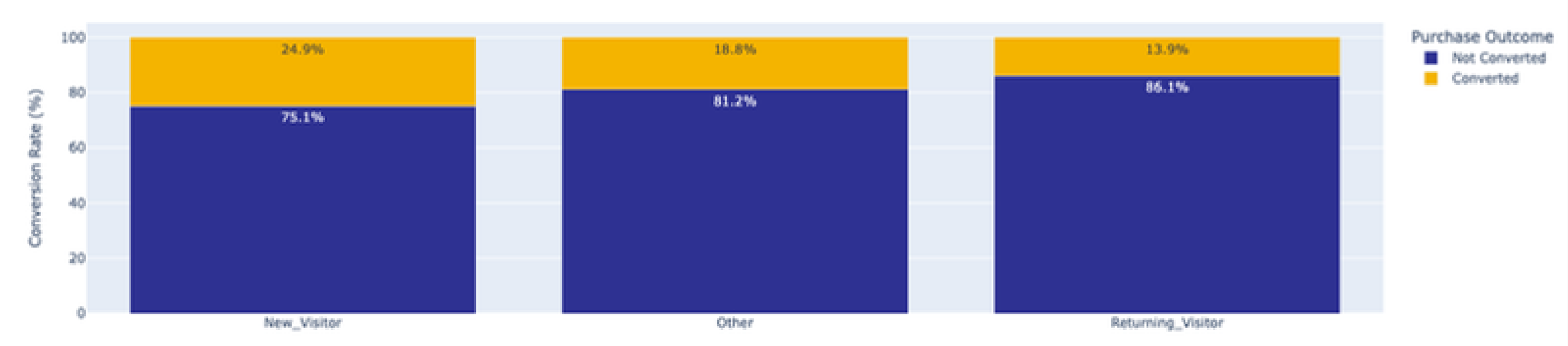
*No session data available for January and April

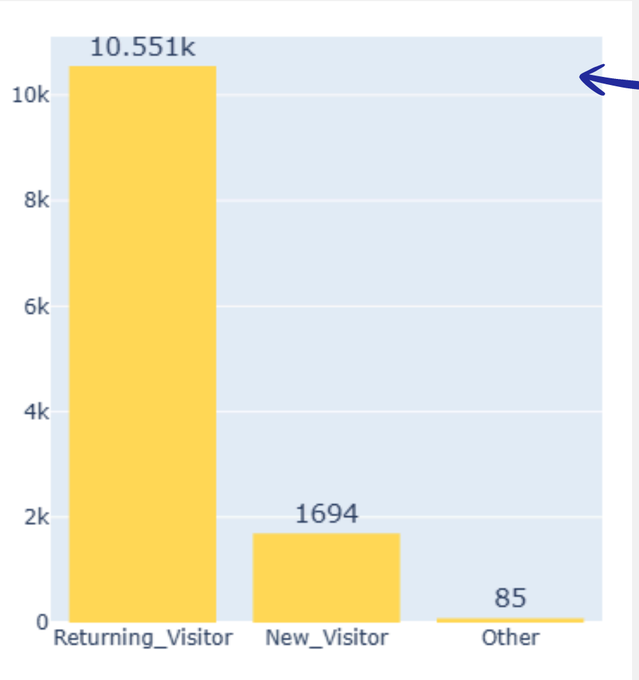
- We see peak acitivity and higher conversions in March, May, November and December. November demonstrates both **high traffic and strong conversion volume**. High traffic months could indicate marketing campaigns driving higher engagement.

- Marketing investment and aggressive targeting can be applied during such high traffic months.

- Holidays like Halloween, Thanksgiving (associated with Black Friday sales) and Christmas could be the driving forces behind this seasonal high spend. But since we do not know the location / year of this dataset, it is difficult to prove this assumption.

Visitor type vs Revenue



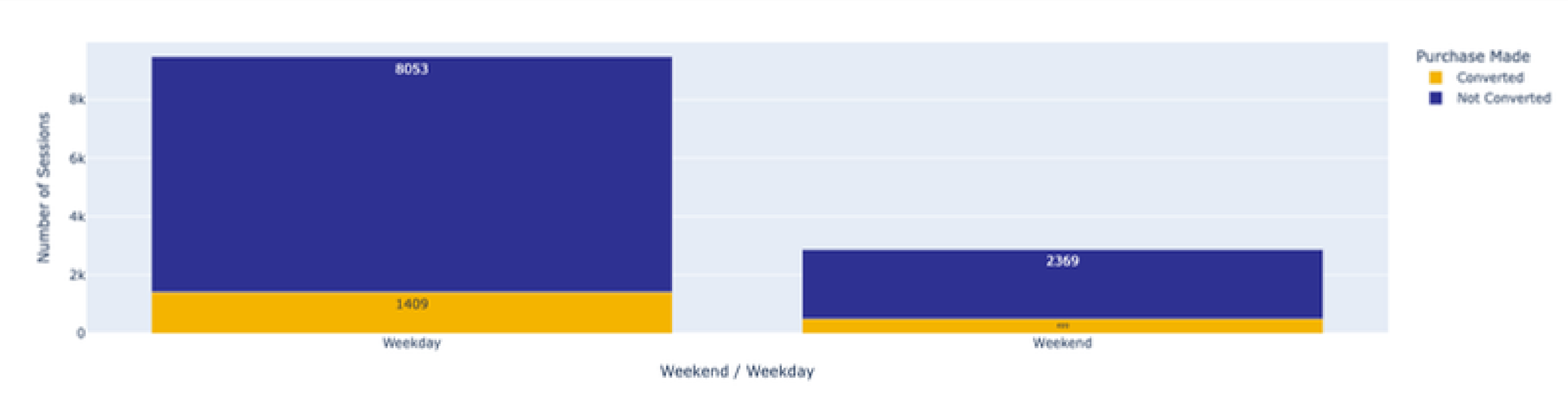


- Visitor Distribution: **Returning visitors** make up the **majority of the audience**, with over 10.5k visits, compared to 1.7k new visitors. Addressing friction points for this group could significantly improve conversions.

- **New visitors have the highest conversion rate**, so first-time visitors are more likely to make a purchase than those who have visited before.This shows a strong-first session intent.

- On the contrary, returning visitors are interested, but not ready to buy which indicates potential friction.

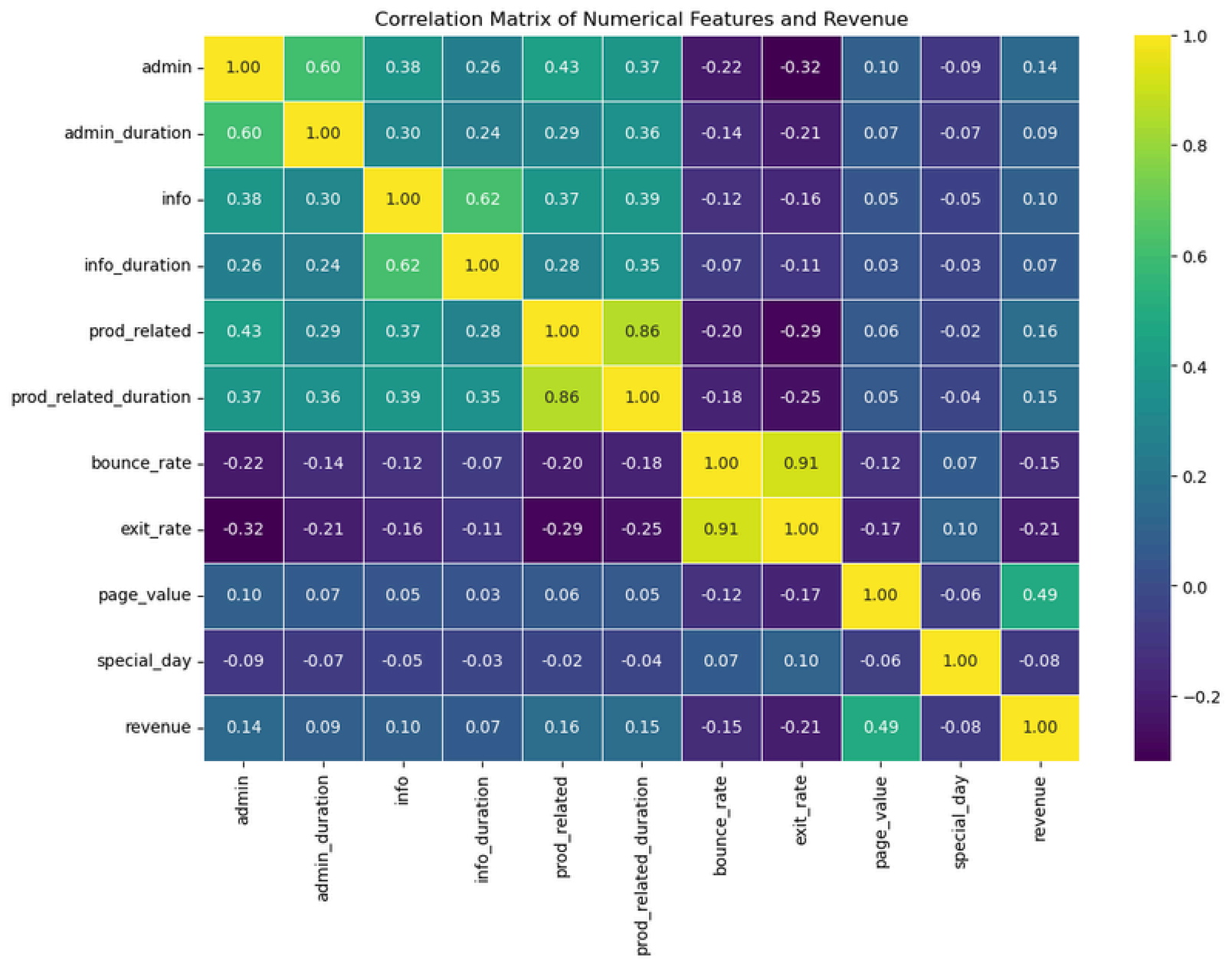
Weekend vs Revenue



- Weekdays → **More traffic, more conversion volume**
- Weekends → **Less traffic, but higher conversion rate**

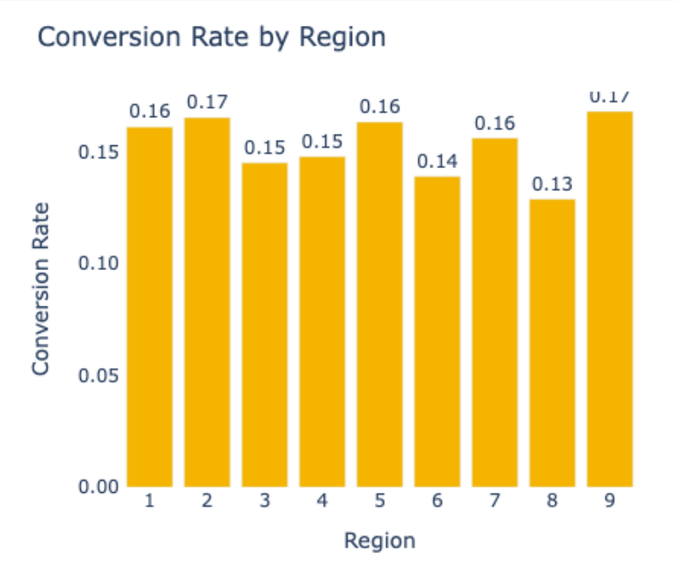
- Weekend visitors have a significantly higher conversion rate than weekday visitors, validated by a z-test, indicating stronger purchase intent on weekends.

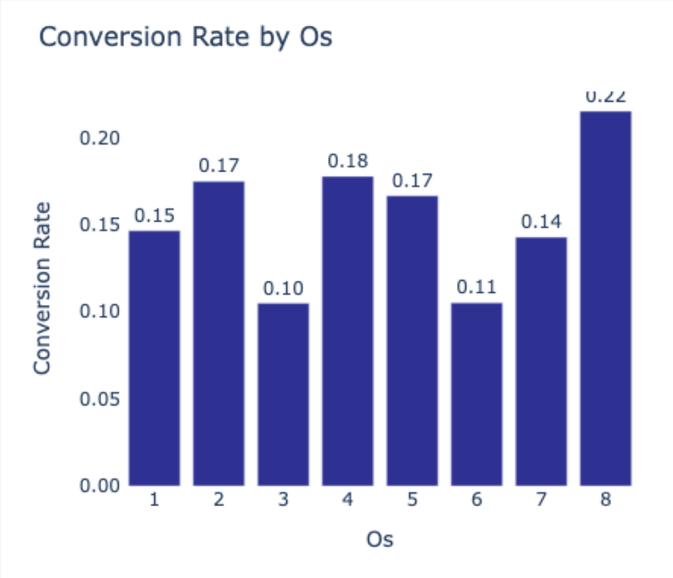
Correlation Matrix

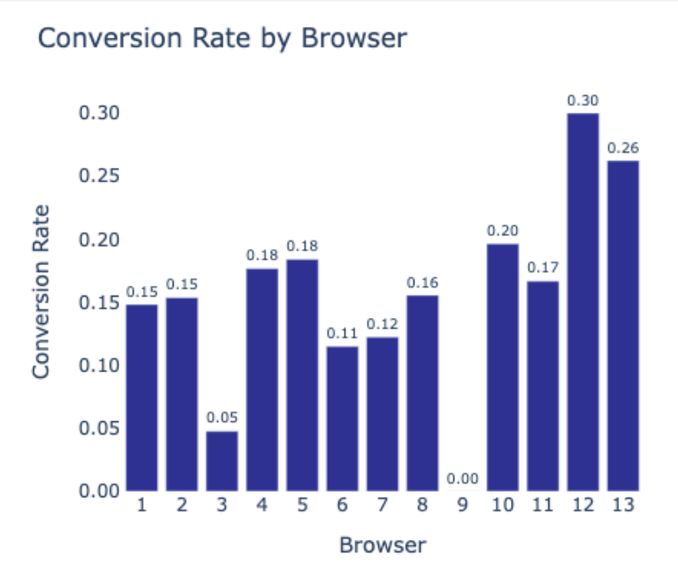


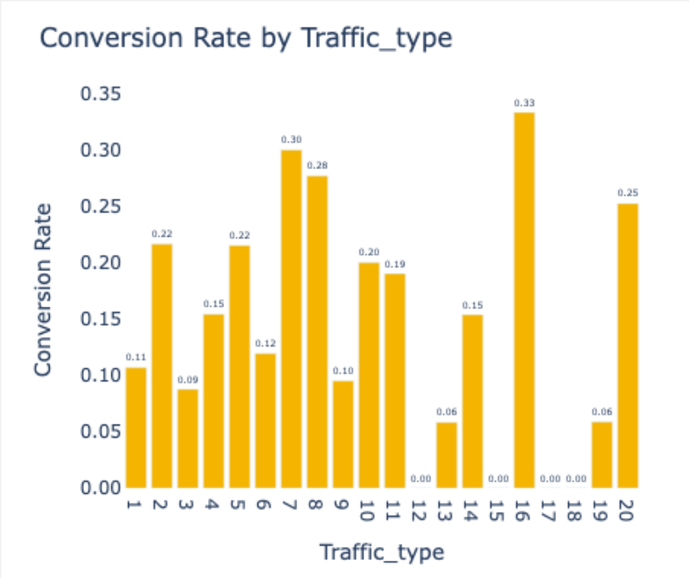
- **Bounce Rate and Exit Rate** move closely together, as both reflect users leaving the website.
- **Exit Rate** shows a **moderate negative relationship with engagement metrics,** indicating that **more engaged users are less likely to leave quickly**.
- **Info and Admin** visits strongly correlate with time spent, since viewing more pages increases session duration.
- **Special Day has very low correlation with other variables,** suggesting user behavior is similar regardless of proximity to special occasions.

Categorical Features vs Conversion Rate









- Conversion rates **vary slightly** across **OS, browsers, regions, and traffic type**, indicating some influence on purchase behavior.
- But these are encoded categories, so they have limited to no interpretation.

Predictive Modeling Approach

Objective

To identify **as many potential buyers as possible** (*high recall*), while **limiting unnecessary targeting** of non-buyers (*balanced precision*), and ensuring **strong overall performance** on imbalanced data (*PR-AUC*)

Model used	Class imbalance handling	Evaluation Metrics
XGBoost Classifier • handles non-linear behaviour • performs well with imbalanced data	Adjusted the **Scale POS Weight** and set it to '3' to ensure model **prioritizes detecting buyers**	• **ROC-AUC** (ranking performance) • **PR-AUC** (performance on imbalanced data) when the metrics above stabilised, • **Recall & Precision** for **buyers**

Model performance

Recall	**72%**
Precision	**64%**
F1-Score	**68%**

- At the selected threshold (~0.6), the model captures 72% of actual buyers.
- With 64% precision, nearly 7 out of 10 targeted users are actual purchasers, indicating efficient targeting while limiting unnecessary outreach

Future model optimisation:

- Evaluate **redundant metrics** (eg- bounce rate and exit rate) for noise reduction
- Explore **feature selection** to simplify the model.
- Categorise the sessions / users into **high, medium and low intent** for closely targeted marketing.
- Replace post-session aggregate metrics with **real-time behavioral indicators** to improve the real-time efficiency of the model.

Recommendations

1.1 Purchase Prediction -

Immediate incentives to encourage checkout

- Limited-time discounts
- Free shipping offers
- Product bundles
- Upsell or premium package recommendations

1.2 No Purchase Prediction -

Engagement and nurturing strategies to increase future conversions

- Social proof (e.g., "X people bought this today")
- Low-stock alerts
- Cart or recently viewed item reminders
- Newsletter signup incentives
- Retargeting ads
- Follow-up email campaigns

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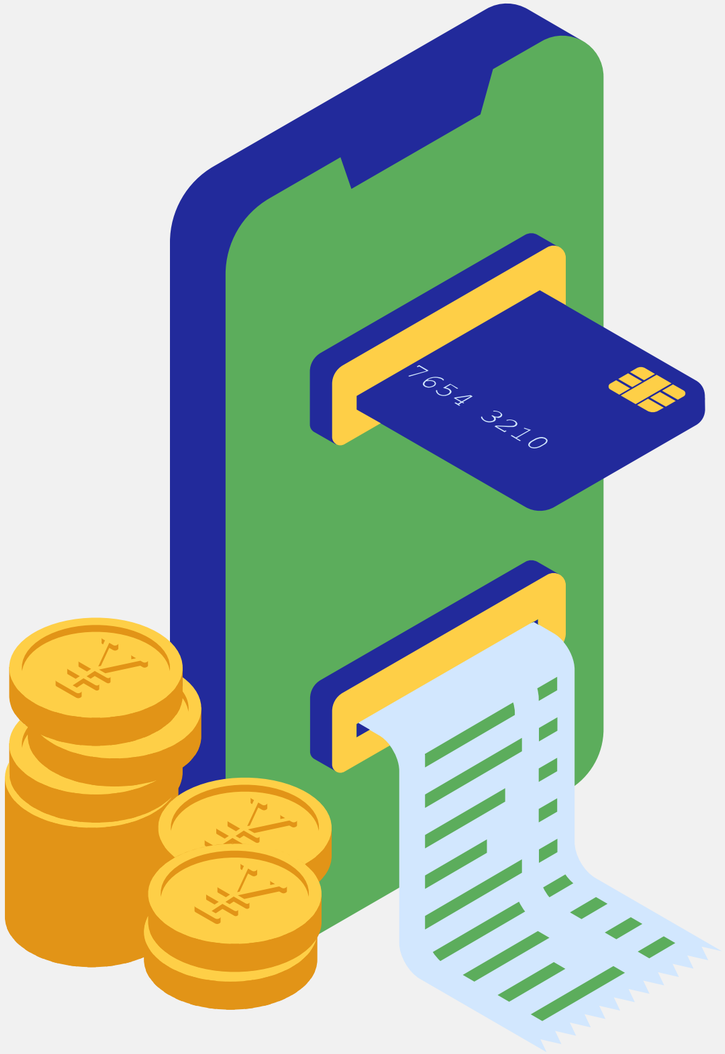
2. Increase targeting aggressiveness during high-conversion months **(e.g., November)**
3. Allocate more promotional effort on **weekends** when purchase intent is higher
4. Optimize the experience for **returning visitors** by addressing possible frictions such as pricing, checkout, availability to encourage them to complete their purchase

LIVE DEMO



THAINK YOU